

Salaheddine Merchich

Software Engineering Student | Java, Spring Boot & React

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Open to international relocation for a 4-6 month final-year internship (PFE) in software development in Europe, available from February 2027 — university internship agreement (EMSI, Morocco).

PROFESSIONAL PROFILE

5th-year engineering student at EMSI Casablanca, with hands-on backend and full-stack experience gained through two internships in industrial environments. Designed and developed EIA SmartFix, a maintenance platform in Spring Boot/React integrating an AI assistant based on RAG (Retrieval-Augmented Generation), and Gestion-de-stock, a stock management platform in React/TypeScript. Comfortable across the full stack — Java/Spring Boot, React, PostgreSQL — with hands-on practice in testing and CI/CD. Seeking a final-year internship (PFE) in software development in Europe.

PROFESSIONAL EXPERIENCE

OCP Jorf Lasfar — Software Engineer Intern

07/2026 - 09/2026

Jorf Lasfar, Morocco • Phosphoric Production Division (EIA) • 4th-year internship

- Designed and developed EIA SmartFix, an industrial maintenance web platform integrating an AI diagnostic assistant based on RAG (Retrieval-Augmented Generation).
- Backend built in Spring Boot following a hexagonal architecture (ports & adapters) for the RAG module; frontend developed in React.
- Implemented hybrid search (dense + lexical) with Reciprocal Rank Fusion (RRF) on PostgreSQL/pgvector (768D embeddings, HNSW indexing), with on-premise LLM inference via Ollama — knowledge base of 118 validated intervention sheets, 21 technical diagrams, and 19 manufacturer manuals, covering 5 equipment families (power supply, instrumentation, mechanics, engine, pump).
- Added a confidence-scoring system for AI-generated answers and orchestrated services via Docker Compose.

Cosumar (SUTA Group) — IT Intern

07/2025 - 08/2025

Oulad Ayad, Béni Mellal-Khénifra, Morocco • IT Department • 3rd-year internship

- Designed and developed Gestion-de-stock, a real-time IT stock management platform (React, TypeScript, Supabase) integrating an AI assistant based on RAG (Ollama), replacing manual paper/Excel tracking.
- Managed and inventoried IT equipment (printers, toners, PC hardware components); organized and cabled the server room.
- Configured and monitored the security camera system; tracked billing and payments via SAP, gaining hands-on understanding of ERP operations in an industrial context.

PROJECTS

Leave-Management-System — Java 17, Spring Boot 3, Angular, JUnit 5, Mockito, Selenium, GitHub Actions

- Leave management platform built with a monorepo architecture (Spring Boot API / Angular frontend), with a complete test suite (unit, integration, E2E) and an automated CI/CD pipeline on GitHub Actions.

Voyage-Zenith — React, TypeScript, Vite, Supabase

- Web platform for booking and organizing trips, with authentication and database powered by Supabase.

Football Arena Quiz — Kotlin/Java, Gradle

- Personal Android football quiz app, built to explore native mobile development outside of web/backend work.

GitHub : github.com/Salaheddine-Merchich • Portfolio : salaheddine-merchich.vercel.app

TECHNICAL SKILLS

Programming Languages : Java, TypeScript, JavaScript, Python, C++

Backend : Spring Boot, REST API, Hexagonal Architecture

Frontend : React, React Native

Databases : PostgreSQL (pgvector), MySQL, SQL Server, Neo4j, Supabase

AI / Data : RAG (Retrieval-Augmented Generation), Vector Embeddings, Vector Search, LLM Inference (Ollama)

Testing & QA : JUnit 5, Mockito, Selenium (E2E), Unit / Functional / Integration Testing, Performance Testing (JMeter), Postman

DevOps & Tools : Docker, Git, GitHub, CI/CD (GitHub Actions)

EDUCATION

EMSI Casablanca — Engineering Degree, Digital Development and Information Systems 09/2022 - 06/2027

State Engineering Degree (Equivalent to Master's / 5-year degree)

High School Diploma, Physical and Chemical Sciences

09/2021 - 06/2022

LANGUAGES

Arabic : Native language

French : B2 (Fluent / Language of instruction)

English : B2 (Professional working proficiency)